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MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

V SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER EXAMINATIONS NOVEMBER 2024

SWITCHGEAR AND PROTECTION [ELE 3123]

Time: 3 Hours **Date: 26 November 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A What is resistance switching? Derive the expression for critical resistance in terms of system inductance and capacitance, which gives no transient oscillation ***(03)***

1B Calculate the RRRV of the 132 kV circuit breaker with neutral earthed. Short-circuit data are as follows: The broken current is symmetrical, the restriking voltage has a frequency of 20 kHz, and the power factor is 0.15. Assume fault is also earthed. ***(03)***

1C Describe with neat sketches the working of the cross-jet explosion pot of an oil circuit breaker. Compare its merits and demerits with other types of arc control devices. ***(04)***

2A What are the different methods of testing of circuit breakers? Discuss their merits and demerits. Explain which method is more suitable for testing the circuit breakers at large capacity. ***(03)***

2B Distinguish symmetrical breaking capacity, asymmetrical breaking capacity, and making capacity, as well as the short-time current rating of a circuit breaker. ***(03)***

2C. With a neat sketch, describe the construction and operation of an HRC Cartridge fuse and why tin joints are used along with silver for the fuse element. ***(04)***

3A. Derive an expression for the Inductance of a Peterson coil for a neutral grounded system.

A 33 kV, 3 Phase, 50 Hz Overhead 80 km line has capacitance to the ground of each line equal to 0.025 μF per km. Determine the Inductance and kVA rating of the Peterson coil. ***(03)***

3B. A 5 A relay is connected to the secondary of CT with a ratio of 300/5 A. The relay operating time is given by t= $\frac{0.15}{(PSM^{0.03})-1}$ *TMS. If the actual operation time is 1.26 seconds for a time multiplier setting of 0.4 when the fault current is 1700 A, determine the current plug setting of the relay. ***(03)***

3C. Discuss the problems associated with the differential protection of a transformer.

With the help of a neat diagram, describe how the Buchholz relay protects the transformer against incipient and short circuit faults. ***(04)***

4A Describe the construction and working of the directional power relay with the help of a neat figure and phasor diagram. Discuss its advantages. ***(03)***

4B Compute the percentage of the stator winding protected against ground faults in a 200MVA, 11kV, 3 phase, 50Hz alternator protected by a current differential protection system. The synchronous reactance of the alternator winding is 35% per phase, and the resistance of the winding is 10% per phase. The protective relay is set to operate when the out-of-balance current exceeds 15% of the full load current. The neutral of the alternator is grounded with a resistance of 2.5Ω.

Also, determine the fault current for a fault in Y phase winding at 95% from the neutral end. ***(04)***

4C Illustrate the 3-zone distance protection of the transmission line with the help of a relevant connection diagram of the relay contacts in the trip circuit of the circuit breaker. Assume that zone-1, and zone-2 employ reactance relay units and zone-3 employs mho relay. Also, show the corresponding combined 3-zone characteristics. ***(03)***

5A Consider two radial feeders protected by over-current relays with very inverse characteristics. Feeder 1 is close to the source (generator). Relay 1 protecting feeder 1 has a plug setting of 125%, and relay 2 protecting feeder 2 has a plug setting of 100%. The time multiplier setting of relay-2 is set to 0.2. Both relay-1 and relay-2 are connected to the secondary of 1000A/5A CT. Select a suitable time multiplier setting for relay-1 so that for a fault current of 9000A in feeder 2, relay-1 provides backup protection. Consider a coordination time of 0.3s. ***(04)***

5B Describe the consequences of an unbalanced supply to a three-phase induction motor. With the help of a relevant circuit schematic explain a suitable protection scheme to protect the induction motor against the consequences of unbalanced supply. ***(03)***

5C With the help of the functional block diagram of a digital/numerical relay, describe the role of the following components

i. Analog input subsystem

ii. Digital input subsystem

iii. Analog/Digital filters

iv. Relay logic implementation ***(04)***